

A Finite Microscopic Latch Reference and Slow-Mode Audit for MQM

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Abstract—An explicit vacuum-bath Hamiltonian supplies a finite-time reference for the MQM completion latch without a Born or Markov approximation. A directed interval calculation certifies retained error below $3/262144 < 2^{-16}$ at $t = 16$ for a 256-mode, finite-band model. A separate strict-resonance control has an exact all-time error floor. For the frozen thirteen-step history clocks, a modal audit identifies weak terminal overlap and slow decay; it does not establish a universal interference obstruction. Continuum certification, finite-temperature operation, physical two-local compilation, and permanent retention remain open.

Index Terms—Quantum control, finite reservoirs, non-Markovian dynamics, autonomous clocks, verified numerics.

I. REQUIREMENTS AND TYPED SCOPE

The Mayes Quantum Method (MQM) control programme requires complete bank and spectator accounting. Its parent uniform history word has a local cold-bath certificate at $1024 \pm 1/1024$. The parent exchange-filter port family fails at $t = 16$. We now address two distinct tasks: an explicit microscopic port reference and a modal diagnosis of the unchanged word generators. Neither is a physical two-local layout.

The reference uses the cold reachable system states $pR0, mR0, eR0, dR1$, denoted $s_0, \dots, s_3$, and the captured state $dR0$. Couplings are $h = g = 1$ and $\lambda = \sqrt{4\sqrt{2}}$. The completion flag is already restricted to its admitted values. A vacuum bosonic bath is new: it is not the earlier thermal Ohmic spectrum. All port operations act as identity on the bank and arbitrary references. This spectator property does not transfer bath conclusions to the complete thirteen-gate word.

Protocols freeze $B \in \{16, 32, 48\}$, $M \in \{256, 512, 1024, 2048\}$, and times 4, 8, 16 before numerical evaluation. Here B is spectral half-width and M counts bath modes, not physical qubits. Exact oscillator-reference dynamics are useful comparison objects; Markov approximations can give qualitatively different populations [1].

II. EXPLICIT FINITE-BATH HAMILTONIAN

Take $\Omega = 64$, $\kappa = 8$, frequencies $\omega_n = \Omega + \delta_n$, and

$$\delta_n = \frac{B(2n+1-M)}{M}, \quad c_n^2 = \frac{\kappa}{2\pi} \frac{2B}{M}, \quad 0 \leq n < M. \quad (1)$$

This is the midpoint discretization of the declared flat spectral density $J(\omega) = \kappa/(2\pi)$ on $\Omega + [-B, B]$. Exchange couples auxiliary excitation $dR1$ to $dR0$ and a bath photon. Total excitation is conserved. The one-excitation subspace, initialized at

TABLE I
FINITE VACUUM REFERENCE AT $t = 16$, $M = 2048$

B	q_M	Distance to local model	Continuum debit
16	1.07125×10^{-5}	6.79784×10^{-6}	1/8
32	6.66356×10^{-6}	2.74313×10^{-6}	1/4
48	5.62358×10^{-6}	1.70166×10^{-6}	3/8

s_0 and bath vacuum, is therefore invariant and has dimension $M + 4$.

In a rotating frame, its exact Hermitian matrix is

$$H_M = \sum_{j=0}^2 w_j (|s_j\rangle\langle s_{j+1}| + \text{h.c.}) + \sum_n \delta_n |b_n\rangle\langle b_n| + \sum_n c_n (|s_3\rangle\langle b_n| + \text{h.c.}), \quad (2)$$

where $(w_0, w_1, w_2) = (1, 1, \lambda)$ and b_n means captured system plus photon n . This finite reference includes coherent frequency shifts and memory directly; no dissipative generator or Lamb-shift subtraction is inserted.

Write the exact state as $\sum_{j=0}^3 v_j(t) |s_j, 0_B\rangle + \sum_n a_n(t) |dR0, 1_n\rangle$. Tracing the original bath gives

$$\rho_S(t) = v(t)v(t)^\dagger \oplus (1 - q_M(t)), \quad q_M(t) = \sum_{j=0}^3 |v_j(t)|^2. \quad (3)$$

The retained error to the captured target is exactly $q_M(t)$. For fixed port launch and arbitrary identity spectators, this also equals retained half-diamond channel error. Photon information is exported to the modeled bath; the bank is not discarded.

Energy distinction. The static full interacting Hamiltonian conserves its total energy. However, if $[H_S, A_\nu] = -\nu A_\nu$, then

$$[H_S + H_B, A_\nu b_\omega^\dagger] = (\omega - \nu) A_\nu b_\omega^\dagger. \quad (4)$$

Thus off-resonant exchange conserves excitation count, not free subsystem energy. Frequency-matched exchange conserves free energy but does not, by itself, produce a Markovian reservoir. No energy-conservation debt is silently removed by changing these definitions.

III. FINITE-TIME VERIFICATION AND CONTINUUM DEBIT

Sparse matrix-exponential action tests all twelve band/grid combinations, with independent dense diagonalization at $M = 256$. Maximum solver-vector disagreement is 1.84×10^{-13} ; grid agreement is a diagnostic, not a continuum certificate. All frozen finite references pass the numerical point target

at $t = 16$. After exposure, the smallest band and smallest mode count were selected for independent interval verification, without tuning their Hamiltonian or deadline. This selection is engineering verification, not an independent prediction.

Proposition 1 (finite reference certificate). For $B = 16$, $M = 256$, bath vacuum, and fixed launch,

$$q_{256}(16) < \frac{3}{262144} < \frac{1}{65536}. \quad (5)$$

Verification. The four-state interaction has norm below four. The bath detuning norm is below sixteen, and its collective coupling obeys $G^2 = 128/\pi < 128/3 < 49$. Hence $\|H_{256}\| < 23 < 24$. A 240-digit directed Taylor calculation through degree 1280 encloses the live amplitudes. For a skew-Hermitian generator, the integral remainder is bounded by

$$\|R_{1280}\| \leq \frac{(24 \times 16)^{1281}}{1281!}. \quad (6)$$

The retained squared norm enclosure is approximately $1.07122284837846 \times 10^{-5}$ and is strictly below the rational ceiling. The exact interval and replay are included. This is a point certificate, not a permanent-hold theorem.

Proposition 2 (continuum comparison). Embed discrete modes as normalized constant functions on equal bins of $L^2[-B, B]$. The flat coupling lies exactly in their span. Replacing continuum frequency multiplication by midpoint steps changes the Hamiltonian by at most B/M in operator norm. Unitary Duhamel and partial-trace contraction therefore give

$$\frac{1}{2}\|\Phi_t^\infty - \Phi_t^M\|_\diamond \leq \min\{1, tB/M\} \quad (7)$$

for the specified fixed-launch spectator channel. Table I shows that this valid bound is too loose to promote the finite pass to the continuum. Continuum quadrature accuracy, finite temperature, and connection to the earlier Ohmic generator remain open.

IV. STRICT-RESONANCE CONTROL: A SCOPED OBSTRUCTION

Collapse all δ_n to zero while keeping total coupling $G^2 = \kappa B/\pi$. Only one collective bright bath mode couples, reducing the dynamics to a closed five-site path with weights $(1, 1, \lambda, G)$. Let $a = \lambda^2/G^2 = \pi\sqrt{2}/(2B)$. Its normalized zero mode is

$$z = \frac{(1, 0, -1, 0, \lambda/G)}{\sqrt{2+a}}. \quad (8)$$

Conservation of $\langle z|\psi_t\rangle$, followed by Cauchy-Schwarz on the four system components, implies $1 \leq \sqrt{2q(t)} + \sqrt{a}$. Therefore

$$q(t) \geq \frac{(1 - \sqrt{a})^2}{2} \quad (a < 1). \quad (9)$$

Using $\pi < 22/7$ and $\sqrt{2} < 99/70$ gives exact all-time floors:

$$B = 16 : \frac{25}{128}; \quad B = 32 : \frac{2209}{8192}; \quad B = 48 : \frac{625}{2048}. \quad (10)$$

This rules out the tested collapsed resonance family at 2^{-16} . It is not a no-go theorem for resonant couplings to other continua or for arbitrary reservoir constructions.

TABLE II
NUMERICAL SLOWEST-MODE DIAGNOSTICS

Profile	Slowest γ	Terminal weight
Uniform	6.96290×10^{-3}	8.70362×10^{-4}
PST16	7.68669×10^{-5}	9.60837×10^{-6}
PST14	1.09334×10^{-4}	1.36668×10^{-5}

V. FROZEN CLOCK DECAY-MODE AUDIT

The parent uniform, PST16, and PST14 profiles retain the same seventeen-state live generator A , launch, λ , and local damping $\kappa = 8$. No weights were retuned. For a normalized right eigenvector $Av = zv$,

$$\gamma = -2 \operatorname{Re} z = 8|v_{16}|^2, \quad (11)$$

because $A + A^\dagger = -8|16\rangle\langle 16|$. Weak terminal support forces slow probability decay. The engineered slowest rates are approximately 91 and 64 times slower. Full biorthogonal expansions reproduce direct propagation within 1.83×10^{-13} over the frozen times. Coefficient squares are not modal probabilities: eigenvectors are not orthogonal and all cross terms are retained. At $t = 1024$, cross terms contribute less than 0.21% of either engineered survival error; long-lived modal components dominate that diagnostic. This supports a weak-terminal-overlap explanation, not a universal destructive-interference claim.

For every positive-weight path in this family, no exact dark eigenmode exists. If $\operatorname{Re} z = 0$, the endpoint identity forces $v_{16} = 0$; the eigen-equation recursively forces all other components to vanish. Every eigenvalue thus has strictly negative real part and $q(t) \rightarrow 0$. These profiles fail the declared deadline, not eventual capture under the local cold generator.

VI. PROMOTION AND NEXT GATE

Evidence separates exact Hamiltonian specifications, derived bounds, finite-reference interval verification, numerical modal diagnostics, and physical realization. Bath grids have phase-alignment scale $\pi M/B$, not a proven full-system recurrence period. A finite bath can re-excite the port and supplies no invariant hold or refreshed reservoir. Physical promotion remains zero.

The next mathematical gate is a tighter continuum propagation enclosure; an orthogonal-polynomial bath-chain representation offers a candidate route. In parallel, extend the reference to a declared finite-temperature supply with explicit truncation and reverse-excitation costs. Neither ideal vacuum success nor a five-bit address embedding closes physical planar two-local compilation or native joint admission.

REFERENCES

- [1] C. Majenz, T. Albash, H.-P. Breuer, and D. A. Lidar, "Coarse-graining can beat the rotating wave approximation in quantum Markovian master equations," *Phys. Rev. A*, vol. 88, p. 012103, 2013.